PRISM: Training-Free, Multimodal Failure Localization for Industrial Manipulation via Topological Persistence

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***Abstract*—State-of-the-art automatic failure-analysis pipelines for robot manipulation are predominantly vision-only: they extract keyframes from pixel motion and hand them to a vision-language model. This is a poor fit for industrial assembly, where many of the failures that matter most — contact jams, over-torqued fasteners, gripper micro-slip, controller saturation — are barely visible on camera but are unambiguous in proprioceptive telemetry. We present PRISM, a training-free, modality-agnostic, zero-shot pipeline that reframes key-event localization as detecting persistent anomalies in a trajectory's raw sensor streams. Per-timestep metrics from any available modality are converted into an empirical-CDF "surprise" score, aggregated across metrics, and reduced with 0-dimensional persistent homology into a small set of scale-invariant candidate events. We evaluate on REASSEMBLE, a real-world, 4551-demonstration industrial (dis)assembly benchmark, and show that a naive Precision@K comparison on this dataset is confounded by extremely wide failure-annotation windows (labeled failures span 69.8% of trajectory duration on average). Stratifying trajectories by failure-duration fraction and reporting a chance-corrected precision score, PRISM shows a statistically significant advantage (Wilcoxon $p \ll 0.05$) over both a uniform-sampling baseline and a pixel-flow (KITE-style) baseline in every cohort, concentrated in force-dominant, visually silent sub-tasks — Insertion and Removal — while showing no advantage on vision-dominant Pick/Place, exactly the scope a multimodal method should claim when force/torque cues are informative. On a separate, hand-annotated set of 50 successful trajectories spanning a taxonomy of salient sub-events (contact, insertion/extraction phases, grasping, stalls), PRISM again leads both baselines on Precision@5 and Recall@5, with its largest margin at tight temporal tolerances ($\leq$1 s), supporting the claim that PRISM generalizes beyond failure detection to salient-event localization more broadly.**

***Index Terms*—failure detection, industrial manipulation, multimodal perception, benchmarking, topological data analysis**

I. Introduction

Robot learning is being asked to meet industrial standards of precision, reliability, and safety, and that bar cannot be met by systems that only report whether a rollout succeeded. Identifying *which moments of a trajectory are salient* — whether they mark a failure, a sub-goal transition, or the climax of a successful maneuver — is a prerequisite for automatic quality control, dataset curation, and teleoperator feedback in deployed manipulation systems, and doing so by hand does not scale to industrial data volumes.

We operationally define a *candidate salient event* as a local maximum of the aggregate rarity score that is unusual relative to the trajectory-wide metric distributions and prominent relative to its temporal surroundings. We hypothesize that these candidates often correspond to task-relevant phenomena, including kinematic failures and large but successful maneuvers; statistical rarity alone, however, does not imply failure.

Existing robot failure and salient-event localization methods are often either trained for a specific setting or rely heavily on visual saliency [1]–[3], [5]. This breaks down in contact-rich manipulation, where the important event may be visible only weakly on camera, yet clearly expressed in proprioception, action, gripper state, force, or motion signals — a component can appear visually unchanged while its joint motion stalls or becomes unusually irregular during insertion. Uniform sampling is, perhaps surprisingly, a strong baseline in this setting whenever ground-truth annotations occupy a large fraction of a trajectory's duration (as we show is the case for REASSEMBLE in §IV), but a high Precision@K score from uniform sampling does not mean it has identified the trajectory's actual event structure — it only reflects the base rate.

We propose PRISM (Persistence-Ranked Identification of Salient Moments), a training-free, multimodal event-localization method that converts heterogeneous signals into empirical-CDF surprise scores and uses topological persistence to rank a sparse set of salient-event candidates without requiring an NMS suppression radius. We evaluate quantitatively on two complementary evaluation settings of REASSEMBLE: REASSEMBLE's failure-annotated trajectories (the primary quantitative result), and a hand-annotated set of 50 successful trajectories where each sub-event is marked with both its temporal range (e.g., 12.9–13.4 s) and its peak timestamp according to rigorous telemetry-based definitions (e.g., 13.02 s where force was highest). Tables III and IV demonstrate that PRISM's advantage extends beyond failure detection to general salient-event localization. Fig. 3 shows a qualitative example of how PRISM, KITE-based, and Uniform select events on a representative REASSEMBLE failure trajectory.

II. Related Work

Existing autonomous failure-analysis and event-localization systems each solve part of this problem.

MATT [1] localizes failures semantically by decomposing explanations into verb (the action that went wrong) and argument (the object/entity involved). However, it provides no temporal or spatial grounding and no quantitative metrics connecting symbolic labels back to raw data.

REFLECT [3] builds hierarchical scene-graph summaries from CLIP-cropped image regions and uses an LLM to pinpoint execution- vs. planning-stage failures. Its main limitation is dependence on known object/asset states and a fixed set of recognized objects; it does not generalize to novel scenes or occluded grasps. Like MATT, it reports success/failure as a binary rather than graded event saliency.

DQAF [4] scores teleoperated trajectories against an assumed expert reference using hand-crafted low-level metrics (joint velocity, acceleration, etc.). This improves interpretability for human operators, but requiring an expert reference sharply reduces generality, and the metric-threshold approach gives only coarse temporal localization (events are regions where a metric exceeds a bound), not pinpoint timing.

AutoEval [5] fuses raw numeric metrics and selected images into a VLM, which outputs a trajectory-level quality score. However, it does not enforce grounding of the VLM's reasoning in the numbers provided, reports a single score per trajectory rather than localized events, and its evaluation is entirely in-distribution.

KITE [2] is a full failure-explanation pipeline: it extracts keyframes via optical-flow magnitude, ranks them by non-maximum suppression (NMS), projects them into a pseudo-bird's-eye-view (BEV) representation, and passes the BEV images to a VLM for semantic explanation of the failure. The core limitation of its temporal selection stage is that NMS's window-size hyperparameter directly determines how many events are detected; varying window size can change the output by an order of magnitude, requiring task-specific tuning. Our evaluation focuses on KITE's keyframe-selection stage alone (pixel-flow + NMS), isolating the temporal localization mechanism from the downstream BEV projection and VLM explanation, which would require architectural changes to adapt to our multimodal input (force/torque signals cannot be meaningfully projected into BEV space).

PRISM builds on two existing tools rather than inventing new machinery. Surprise-scoring adapts ECOD [7], which computes per-dimension left/right-tail empirical-CDF scores and sums them across dimensions to obtain one outlier score per point, assuming i.i.d. samples from a fixed population. We repurpose this to score timesteps within a single trajectory against that trajectory's own values — treating time as the reference population rather than a dataset of i.i.d. points, a setting ECOD's convergence guarantees were not designed for (discussed in §III-B). For event selection, we replace ECOD's native threshold/top-K cutoff with 0-dimensional persistent homology [8], an established technique for noise-robust peak ranking, equivalent to topographic prominence [9], [10], [11], and previously applied to structure detection in generic time series [12]. Our contribution is this specific pairing — ECOD-style per-trajectory scoring selected via persistence rather than thresholding — applied to multimodal robot telemetry, not a new scoring rule or a new topological construction.

Each existing method is either trained for a specific setting, evaluated only in-distribution, or dependent on visual saliency; none provide a training-free, scale-invariant, parameter-light way to select candidate events directly from raw, multimodal telemetry — the gap PRISM addresses.

III. Method: PRISM

A. Multimodal metric extraction

For every trajectory we compute a bank of per-timestep scalar signals from the modalities available in the logged data. On REASSEMBLE, these include:

Let $\mathbf{q}_i \in \mathbb{R}^d$ and $\dot{\mathbf{q}}_i \in \mathbb{R}^d$ denote the measured joint positions and joint velocities at timestamp t_i, respectively; $\mathbf{u}_i$ denotes the commanded action vector, $\mathbf{g}_i$ the measured gripper position, and I_i the RGB image. The metric bank contains:

1) *Motion smoothness* is represented by the norm of joint-space jerk. We first estimate velocity, acceleration, and jerk by finite differences:

$$\widehat{\dot{\mathbf{q}}}_i = \frac{\mathbf{q}_i - \mathbf{q}_{i-1}}{t_i - t_{i-1}}, \tag{1}$$

$$\widehat{\ddot{\mathbf{q}}}_i = \frac{\widehat{\dot{\mathbf{q}}}_i - \widehat{\dot{\mathbf{q}}}_{i-1}}{t_i - t_{i-1}}, \tag{2}$$

$$m_i^{\text{jerk}} = \left\| \frac{\widehat{\ddot{\mathbf{q}}}_i - \widehat{\ddot{\mathbf{q}}}_{i-1}}{t_i - t_{i-1}} \right\|_2 . \tag{3}$$

This metric is calculated from measured joint positions, not end-effector velocity.

2) *Velocity magnitude* measures the norm of the recorded joint-velocity vector:

$$m_i^{\text{vel}} = \|\dot{\mathbf{q}}_i\|_2 . \tag{4}$$

Unusually large values are interpreted as velocity spikes by the subsequent rarity-scoring stage. This is a joint-space metric, not an end-effector metric.

3) *Joint-speed magnitude* measures motion between consecutive joint configurations:

$$m_i^{\text{speed}} = \frac{\|\mathbf{q}_i - \mathbf{q}_{i-1}\|_2}{t_i - t_{i-1}}. \tag{5}$$

Unusually low values are interpreted as candidate stalls. The implementation does not use a platform-specific velocity threshold or verify that commanded motion continued during the stall. This is calculated in joint space.

4) *Optical-flow magnitude* is computed from dense Farnebäck optical flow $\mathbf{F}_i$ between consecutive RGB frames:

$$m_i^{\text{flow}} = \frac{1}{|\Omega|} \sum_{\mathbf{p} \in \Omega} \|\mathbf{F}_i(\mathbf{p})\|_2 , \tag{6}$$

where Ω denotes the evaluated image pixels. This is a visual metric and does not use joint or end-effector state.

All signals are computed at the data's native sample rate, then aligned to a common timestep. These signals are treated uniformly as scalar time series by the downstream surprise-scoring and event-selection stages, so PRISM degrades gracefully to whatever sensors a platform logs rather than requiring a fixed sensor suite — a robot with only IMU and gripper state could use a subset of these; a system with additional tactile sensors could add additional metrics without architectural changes. In addition to the metrics used for ECOD scoring, we record gripper open–close transitions as auxiliary semantic markers.

B. ECDF-based surprise aggregation

Adapting the ECOD outlier-scoring framework [7] to trajectory data, we compute per-metric surprise scores that capture how unusual each timestep is relative to the trajectory's own distribution. The intuition is simple: important trajectory events may produce values near the tails of a metric's empirical distribution—either unusually high, such as large joint jerk during an abrupt maneuver, or unusually low, such as near-zero joint speed during a stall. We therefore evaluate both tails of each metric without assuming that only large values are informative.

For metric j with trajectory values $X_1^{(j)}, \ldots, X_n^{(j)}$, we compute the empirical left- and right-tail probabilities

$$F_{\text{left}}^{(j)}(z) = \frac{1}{n} \sum_{i=1}^{n} \mathbb{1}\left\{ X_i^{(j)} \leq z \right\}, \tag{7}$$

$$F_{\text{right}}^{(j)}(z) = \frac{1}{n} \sum_{i=1}^{n} \mathbb{1}\left\{ X_i^{(j)} \geq z \right\}. \tag{8}$$

The corresponding negative-log tail scores at timestep t are

$$U_{tj}^{L} = -\log\left(\max\left\{ F_{\text{left}}^{(j)}(X_t^{(j)}), \epsilon \right\} \right), \tag{9}$$

$$U_{tj}^{R} = -\log\left(\max\left\{ F_{\text{right}}^{(j)}(X_t^{(j)}), \epsilon \right\} \right), \tag{10}$$

where $\epsilon > 0$ provides numerical stability. Following ECOD, we additionally select a tail according to the empirical skewness of each metric,

$$\gamma_j = \frac{\frac{1}{n} \sum_{i=1}^{n} \left(X_i^{(j)} - \bar{X}^{(j)} \right)^3}{\left[\frac{1}{n} \sum_{i=1}^{n} \left(X_i^{(j)} - \bar{X}^{(j)} \right)^2 \right]^{3/2}}, \qquad s_j = \operatorname{sgn}(\gamma_j). \tag{11}$$

The skew-selected score is

$$U_{tj}^{S} = -\operatorname{sgn}(s_j - 1)\, U_{tj}^{L} + \operatorname{sgn}(s_j + 1)\, U_{tj}^{R}. \tag{12}$$

Thus, $U_{tj}^{S} = U_{tj}^{R}$ for positively skewed metrics, $U_{tj}^{S} = U_{tj}^{L}$ for negatively skewed metrics, and $U_{tj}^{S} = U_{tj}^{L} + U_{tj}^{R}$ when the measured skewness is zero. The aggregate trajectory rarity score is then

$$O_t = \max\left(\sum_j U_{tj}^{L}, \sum_j U_{tj}^{R}, \sum_j U_{tj}^{S} \right). \tag{13}$$

This aggregation differs from independently taking the maximum tail score for every metric before summation: it first accumulates evidence consistently within each tail and then selects the largest aggregate tail response. For Stalling we always take the left tail as a design choice.

Scores are summed across metrics, $O_t = \sum_j O_t^{(j)}$, and smoothed before persistence is computed. The results in §IV use a centered Savitzky–Golay filter with an effective window of 11 samples and polynomial order three. Within each window, the filter estimates the center value by fitting a low-degree polynomial using least squares. For a fixed window and polynomial order on an equally sampled signal, this operation is equivalent to convolution with a fixed set of coefficients; its advantage is therefore not adaptation of the kernel, but preservation of local polynomial trends. Compared with a box filter, it generally produces less flattening and fewer plateau artifacts, while compared with Gaussian smoothing it can better preserve the height and width of narrow peaks for the parameters used here. This comes without the Gaussian scale-space guarantee: Savitzky–Golay filtering can overshoot, introduce additional extrema, and produce small negative values. We clamp negative values to zero before computing persistence. We selected this filter empirically because it gave better localization performance in our evaluation, rather than claiming that it is universally superior to Gaussian or box smoothing. We are explicit that O_t is a within-trajectory rarity heuristic, not a calibrated probability: the underlying process is neither i.i.d. nor stationary, so the Glivenko–Cantelli convergence guarantee behind the ECDF does not hold here, and summing across correlated metrics does not assume independence — it is a deliberate corroboration mechanism, since two metrics agreeing on an anomaly should reinforce it. The trade-off is that redundant metrics can inflate a peak; we treat this as a limitation mitigated by favoring modality diversity over metric count, not a correctness issue.

C. Topological event selection

NMS requires an arbitrary window-size hyperparameter, and changes the number of detected events as that hyperparameter changes. We instead rank candidates using 0-dimensional persistence — equivalent to topographic prominence [9], [11] — applied here to the aggregated multimodal surprise score O_t rather than a single raw signal. In this framework, each local maximum is the birth of a connected super-level-set component, which merges into a taller neighbor at a local minimum; persistence is defined as the birth height minus the merge (death) height. We rank candidate events directly by this persistence value, so an event is scored by how far it stands above its own surrounding local structure, not against a single global cutoff.

In earlier work, we experimented with ranking events by the area under each event's local basin, intended to down-weight instantaneous, single-timestep sensor-glitch spikes relative to sustained anomalies. Ranking by persistence height alone gave marginally higher precision (detailed results not shown), so we adopt it as the primary, simpler criterion and report it

throughout §IV. One trade-off of this simplification is that persistence height does not distinguish a genuinely sustained anomaly from a brief spike of the same amplitude; we rely on the smoothing stage (§III-B) to remove non-physical, single-sample spikes before persistence is computed, rather than encoding that distinction into the ranking rule itself. Importantly, ranking by persistence height retains the ℓ_∞ stability guarantee: a bounded uniform perturbation of O_t bounds how much the ranking can be reordered, a property that area-based ranking does not enjoy.

IV. Evaluation on REASSEMBLE

A. Dataset and the failure-window bloat problem

REASSEMBLE [6] is a real-world, multimodal, contact-rich industrial assembly/disassembly dataset built on the NIST Assembly Task Board #1: 4,551 high-level demonstrations across four task families — Pick, Insert, Remove, Place — of small industrial parts, with synchronized RGB, force/torque, audio, and proprioceptive streams; 516 high-level demonstrations are labeled failures, each with a start/end timestamp. We select the top $K{=}5$ events per trajectory and report Precision@5 (the fraction of the top-5 selected events that fall inside the labeled failure window). The choice of $K{=}5$ is a practical one: for a 30–60 second manipulation trajectory, selecting 5 salient events corresponds to roughly one critical moment per 6–12 seconds of real-world robot time, which is a useful granularity for both human review and downstream automation (e.g., framing a VLM query with 5 candidate images rather than sifting through 1,500 frames).

A naive Precision@5 comparison on the full dataset is confounded by how failures are annotated: a labeled failure window spans, on average, 69.8% of its trajectory's duration, so a selector that ignores the signal entirely already scores well by construction — Uniform sampling alone reaches 69.2% precision on the unfiltered dataset (Table I, "Full" column). This makes the full-dataset comparison a weak test of whether a method is actually localizing anything, and it is the reason we do not treat it as our primary result. We address this in two ways. First, we stratify trajectories into cohorts by the fraction of their duration labeled as failure ($<$20%, $<$30%, $<$40%, $<$50%, $<$60%, and the unfiltered Full set) and report Precision@5 within each cohort, so every method faces the same base rate. Second, within each cohort we report a chance-corrected precision,

$$\text{Adjusted} = \frac{P@5 - \bar{c}}{1 - \bar{c}}, \tag{14}$$

where $\bar{c}$ is the cohort's mean failure-duration fraction, i.e. the expected precision of a selector with no information about the signal. This rescales the raw gap over Uniform by how much headroom above chance was actually available, so a method cannot look skillful merely by being evaluated on trajectories with a high base rate. We compare against Uniform sampling and a *KITE-based baseline* that re-implements only KITE's pixel-flow-plus-NMS temporal front end for keyframe selection, excluding the downstream pseudo-bird's-eye-view projection and VLM explanation stages. This isolates the event-selection mechanism for fair comparison, since KITE's full pipeline includes components (BEV, VLM) that are inherently vision-centric and difficult to extend to force/torque signals. Optical flow parameters (resize: 128×128, frame-stride: 2, NMS radius: 5) were selected for computational efficiency; we verified that larger resolutions (e.g., 1024×1024) did not improve event selection quality at the cost of significantly increased runtime.

B. Results

Table I shows PRISM's precision exceeding both baselines in every cohort, including the unfiltered Full set. Uniform's own adjusted score stays within $\pm$2.6 points of zero across all six cohorts, as it should — Uniform's expected precision is the base rate by definition, so a near-zero adjusted score is a sanity check that our correction is calibrated, not a result in itself. PRISM's adjusted score is positive and substantially larger everywhere, from 3.4 points at the tightest cohort to 9–11 points at the looser cohorts and on the full dataset, and paired Wilcoxon signed-rank tests reject the null of no difference from Uniform at $p \ll 0.05$ in every cohort.

The raw precision gap follows a mild hump shape — smallest at $<$20% (a floor effect: with a low base rate and a fixed budget of $K{=}5$, no method can push precision far above the base rate) and at Full (a ceiling effect: both methods are already near 70–75%) — but the chance-corrected score shows the advantage is not an artifact of either end of that range. The absolute precision numbers in tighter cohorts ($<$50%, $<$60%) are modest (29–40%) because PRISM, as a salient-event localizer, selects one peak per structurally significant event in a trajectory; failure trajectories in REASSEMBLE can contain multiple such salient events, most of which may not lie in the failure region, so with a budget of $K{=}5$, coverage is selective. This is consistent with the 70.0% Recall@5 on the 50 hand-annotated successful trajectories (Table III), which shows that PRISM reliably places at least one peak in every major salient moment, whether that moment is a failure or a nominal sub-event. The fact that PRISM's capability transfers from failures to successful-trajectory peaks further supports the interpretation that PRISM is detecting a general signal structure, of which failure is one instance.

C. Per-task breakdown

Table II breaks Precision@5 down by task, both within the $<$50% cohort and on the full, unfiltered dataset.

The pattern that motivated this method holds in both views: PRISM leads Uniform by 5.1 points on Insert and 14.8 points on Remove in the $<$50% cohort (4.9 and 6.0 points, respectively, on the full dataset) — the two task families where force and torque cues are dominant and carry information that pixel motion alone cannot capture. On the two vision-dominant tasks (Pick and Place), PRISM is as good as Uniform: on the full, unfiltered dataset, Pick achieves 91.4% for PRISM vs. 91.3% for Uniform, and Place 84.3% vs. 84.4%, essentially equivalent performance. In the $<$50% cohort specifically, Pick

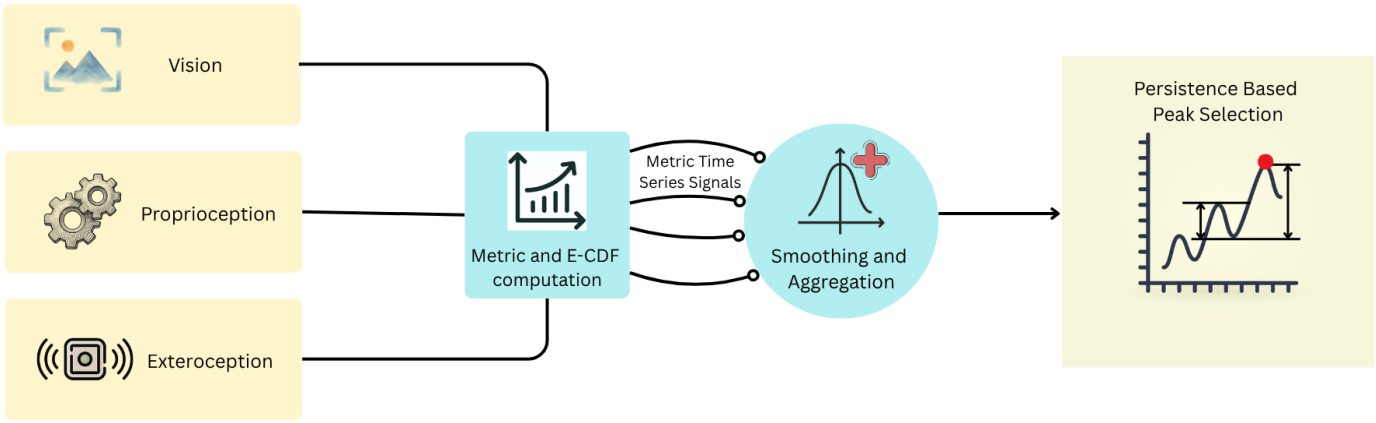


Fig. 1. PRISM pipeline: multimodal metrics → ECDF surprise scores → aggregation and smoothing (depicted as Gaussian for visualization; Savitzky–Golay is utilized in evaluation) → persistence-ranked event selection. The output is a ranked list of candidate failure/anomaly timestamps; downstream semantic labeling (e.g. VLM captioning of selected events) is future work and is not evaluated here.

TABLE I
PRECISION@5 ON REASSEMBLE, STRATIFIED BY FAILURE-DURATION FRACTION (MEAN [95% BOOTSTRAP CI], %), AND CHANCE-CORRECTED PRECISION

	<20%	<30%	<40%	<50%	<60%	Full
Precision@5						
Uniform	14.2 [11.6, 16.9]	20.7 [18.2, 23.2]	26.0 [23.9, 28.2]	29.7 [27.7, 31.5]	32.6 [30.5, 34.7]	69.2 [66.2, 72.2]
KITE-based	10.7 [7.1, 14.7]	10.3 [7.8, 13.1]	11.8 [9.5, 14.3]	12.9 [10.7, 15.2]	12.8 [10.7, 15.1]	57.0 [53.0, 61.0]
PRISM (ours)	**18.2 [14.2, 22.2]**	**25.3 [21.4, 29.4]**	**31.3 [27.4, 35.2]**	**38.6 [34.9, 42.3]**	**41.9 [38.3, 45.6]**	**74.3 [71.4, 77.2]**
Chance-corrected precision						
Mean failure fraction $\bar{c}$	13.5	19.0	24.2	29.8	32.4	69.8
Uniform, Adjusted	0.8	2.1	2.4	−0.3	0.3	−2.2
PRISM, Adjusted	**5.4**	**7.8**	**9.3**	**12.5**	**14.1**	**14.8**

Bold = best per column. Cohorts are cumulative (e.g. "<50%" = all trajectories with failure fraction below 50%). CIs from bootstrap resampling of trajectories within each cohort (10,000 resamples).

TABLE II
PRECISION@5 (%) BY TASK CATEGORY ON REASSEMBLE

Task	$n_{<50\%}$	Unif.$_{<50\%}$	PRISM$_{<50\%}$	Unif. Full	PRISM Full
Pick	11	25.5	25.5	91.9	91.9
Place	6	40.0	26.7	84.3	80.9
Insert	115	28.5	**36.9**	53.0	**60.5**
Remove	42	32.4	**48.6**	74.5	**81.1**

$n_{<50\%}$ = trajectories in the <50% failure-fraction cohort. Bold = PRISM ahead by >1 point; Pick/Place have too few trajectories in that cohort for either direction to be meaningful.

shows a 3.7-point dip (21.8% vs. 25.5%), but this cohort contains only 11 Pick trajectories, rendering this small sample highly susceptible to noise; the full-dataset equivalence is the reliable signal. We read this as evidence that PRISM's advantage lies in multimodal integration: when force/torque signals are more informative than visual motion, PRISM recovers this structure; when vision is already a strong signal, adding force/torque metrics provides no loss in performance. This is exactly the modality-agnostic behavior we claim in §I — not a rejection of vision, but an approach that lets the data tell which sensors are informative, rather than privileging any single modality a priori.

D. Per-metric importance

To understand which modalities drive PRISM's performance, we ablated each metric individually on the <60% cohort and the full cohort Fig. 2. Removing the stalling signal (force-based contact detection) reduced point precision@5 from 39.8% to 37.8%, and removing velocity-spike detection reduced it to 38.4% on the <60% cohort, confirming that force/torque metrics are the primary drivers of PRISM's advantage. Conversely, removing optical flow had negligible impact (39.9%), and removing action discontinuity produced identical results, suggesting these modalities carry little signal for this task. Similar but less pronounced trend was seen for the full cohort. This empirically validates the multimodal approach: PRISM automatically assigns weight to whichever signals are informative for the task, rather than privileging vision a priori.

E. Salient-event localization on successful trajectories

To validate that PRISM's advantage extends beyond failure detection to general salient-event localization, we hand-annotated 50 successful REASSEMBLE trajectories using a taxonomy of sub-events that occur during nominal task execution. The annotation protocol defines each event precisely, with explicit reference to both visual and telemetry cues:

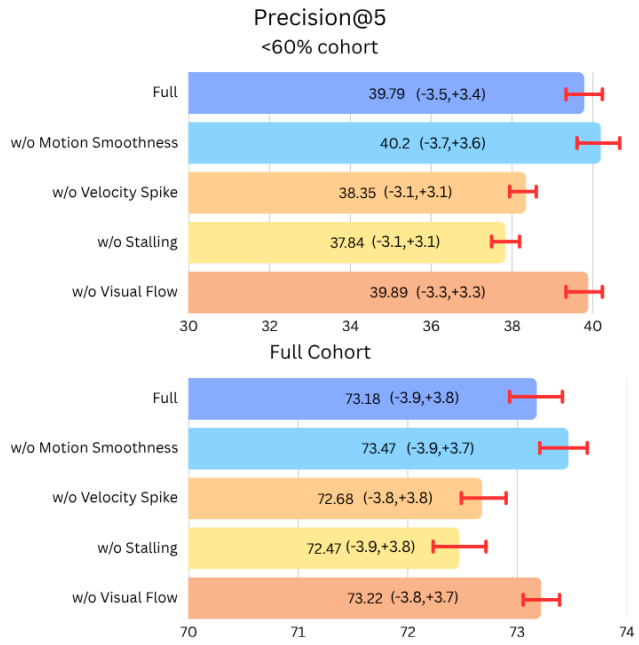


Fig. 2. Per-metric ablation reveals that force-based signals (stalling, velocity spike) are the primary drivers of PRISM's performance on REASSEMBLE, while visual flow has minimal impact

Approach: reaching motion from current pose toward the task region, spanning a temporal range (e.g., 2–4 s as identified from video); the salient moment is marked by the highest-velocity timestep during this reaching motion (e.g., peak at 2.4 s per joint/end-effector velocity telemetry).

Contact: first timestep where the end-effector or held object makes physical contact with a target object or surface.

Engaged rotation: rotating while in contact with an object; typically produces sustained high-amplitude values in force/torque telemetry.

Active push/pull: visible movement of an object or gripper; marked by the highest-force timestep during the push/pull region (from force/torque signals). Pull events decompose into (a) engaged rotation, (b) active pull (visual motion), and (c) release (gripper open).

Insertion: composed of engaged rotation (rotating while pushing inward) and active push with inward motion constraint; defines sub-goals such as threading or seating a fastener.

Gripper open/close: timestep when gripper state transitions from closed to open or vice versa.

Stall: visually apparent momentary pause in motion despite continuing commands.

Events can overlap temporally — for example, an insertion simultaneously involves engaged rotation and active push — and we record both the temporal range (start/end times from video review) and the peak timestamp (identified via highest telemetry value) for each annotated event. The annotation was performed by the authors using frame-by-frame video review and careful reference to force/velocity telemetry signals to identify event ranges and peaks according to the rigorous definitions above.

TABLE III
PRECISION@5 AND RECALL@5 ON 50 HAND-ANNOTATED SUCCESSFUL TRAJECTORIES

Selector	Precision@5	95% CI	Recall@5	95% CI
Uniform	21.6%	[16.4, 26.8]	48.9%	[37.8, 61.0]
KITE-based	34.8%	[25.2, 44.8]	38.9%	[29.4, 48.8]
PRISM	**39.2%**	**[31.6, 47.2]**	**70.0%**	**[59.6, 81.0]**

TABLE IV
TEMPORAL PRECISION (%) ON SUCCESSFUL TRAJECTORIES: FRACTION OF TOP-5 EVENTS WITHIN ERROR TOLERANCE OF ANNOTATION

Selector	≤ 1 s	95% CI	≤ 2 s	95% CI	≤ 5 s	95% CI
Uniform	27.7%	[19.4, 36.4]	73.1%	[63.4, 82.7]	96.9%	[92.9, 100.0]
KITE-based	30.8%	[23.1, 38.7]	40.0%	[31.9, 48.7]	66.2%	[57.5, 74.8]
PRISM	**56.2%**	**[45.6, 67.2]**	72.3%	[63.0, 81.5]	91.5%	[86.2, 96.2]

Table III reports Precision@5 and Recall@5 on this curated set, while Table IV reports temporal precision at increasingly strict error tolerances (≤ 1 s, ≤ 2 s, ≤ 5 s), measuring what fraction of selected events lie within each window of an annotated event. PRISM leads both baselines on Precision@5 (39.2% vs. 34.8% for KITE-based, 21.6% for Uniform) and especially on Recall@5 (70.0% vs. 38.9% for KITE-based, 48.9% for Uniform), indicating it recovers the trajectory structure better. The temporal-precision results show the advantage is sharpest at tight tolerances: PRISM reaches 56.2% within ≤ 1 s versus 30.8% for KITE-based and 27.7% for Uniform. At looser tolerances (≤ 2 s and ≤ 5 s), all methods converge, suggesting most trajectories' salient events cluster within 2–5 second windows of each other. Given that trajectories range from 10–50 s in duration, a 2–5 s event window represents a substantial fraction of a trajectory, and the fact that all methods achieve 40–97% precision in this band indicates that identifying "a region where something interesting happened" is relatively easy. The real discriminator is precision at sub-second tolerances (≤ 1 s), where PRISM reaches 56.2%, showing it recovers the climax or peak moment of each salient event — the precise instant when the event's defining characteristic (highest force, highest velocity, etc.) occurs according to the telemetry. This capability is exactly what the pipeline is optimized for: detecting and ranking events by their telemetry-defined peaks, which are the moments an automated system should flag for human review or downstream processing.

Fig. 3 shows the frames selected by each method on a representative trajectory: the top row is PRISM's top-5 events, the middle row is KITE-based, the bottom row is Uniform, and the bottom panel is the full-trajectory aggregate O_t curve with all three methods' picks marked. PRISM's selections align with physically salient moments (rapid motion, high force) whereas KITE-based picks events of maximum optical flow regardless of contact dynamics, and Uniform samples uniformly across time.

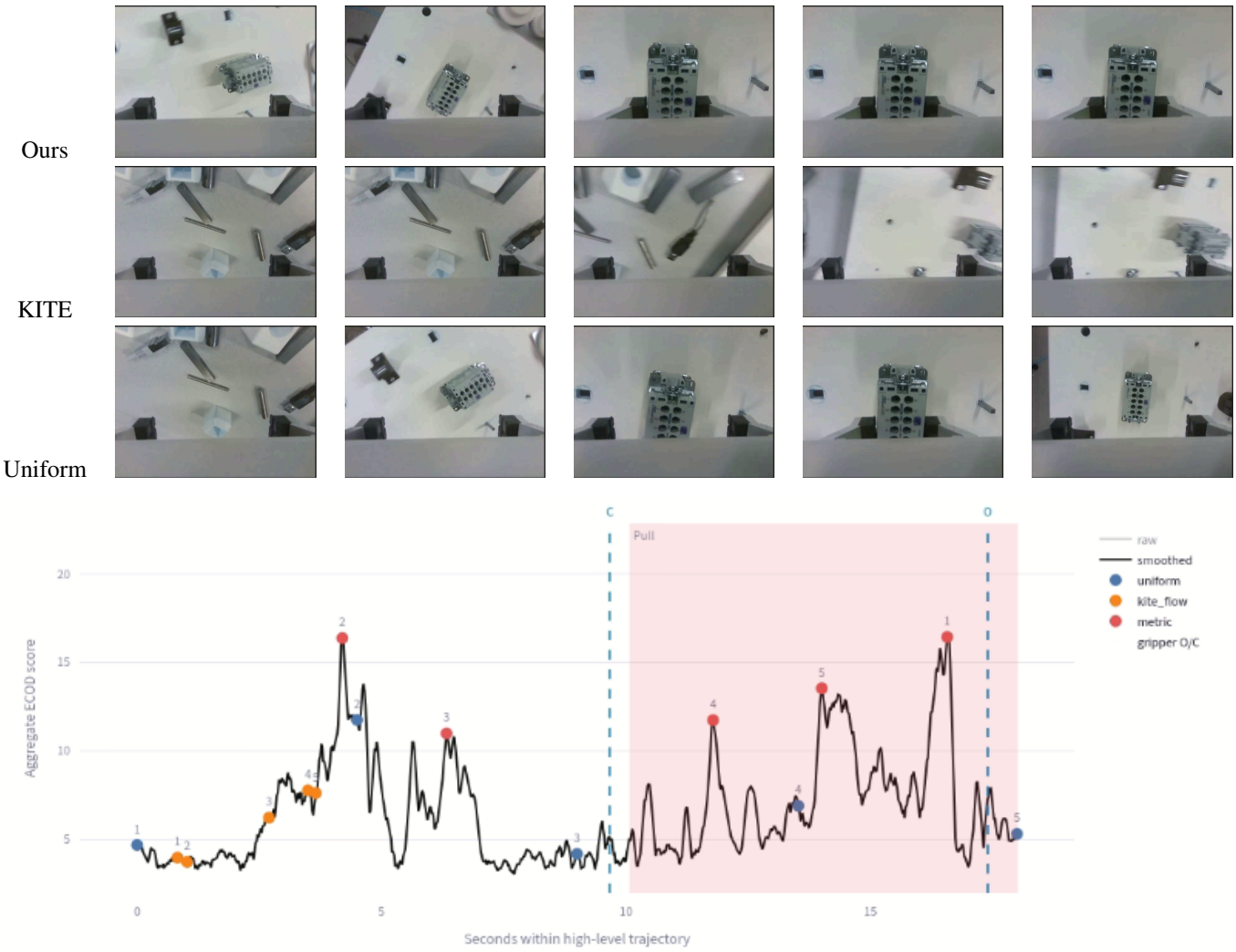


Fig. 3. Qualitative comparison of top-5 selected events on a representative REASSEMBLE trajectory (shaded pink region marks the labeled failure window). Rows, top to bottom: PRISM (ours), KITE-based, Uniform. Bottom: aggregate surprise score for the full trajectory with each method's picks marked. Images are ordered left to right by increasing time. KITE-based selects events where visual optical flow is maximal, but completely misses out on failures which are better diagnosed using telemetry, while Uniform samples uniformly across time. PRISM selects anomalies that are large relative to their local context; on this trajectory, these correspond to physically salient moments: the rapid approach phase and the high-force pull-and-extraction phases where the arm applies significant force to the object

V. Discussion and Limitations

Metric redundancy. PRISM's aggregation sums per-metric surprise scores, so redundant metrics that measure near-identical quantities can jointly inflate a single peak. We treat corroboration between independent evidence sources as a feature (two metrics agreeing on an anomaly should reinforce it) but acknowledge this as a limitation, best mitigated by favoring modality diversity over metric count rather than by any change to the scoring rule. Adding metrics that measure orthogonal phenomena (e.g., tactile, audio) rather than near-duplicates of existing joint/force signals would be the principled fix.

Vision-dominant tasks. On the full, unfiltered dataset, PRISM performs equivalently to Uniform on both Pick and Place (91.4% vs. 91.3%, and 84.3% vs. 84.4%, respectively), confirming that PRISM does not degrade performance on vision-dominant tasks. The $<$50% cohort shows Pick at 21.8% for PRISM vs. 25.5% for Uniform, but this cohort contains only 11 Pick trajectories, making it susceptible to noise and sample-specific variation. The reliable conclusion from the full dataset is that PRISM's multimodal integration is a strict improvement for force/torque-heavy sub-tasks (Insert, Remove) and is neutral for vision-dominant tasks — exactly what we expect from a method that aggregates both modalities and lets the data determine which signals carry information.

Future work. Spatial grounding (bounding box prediction for where in the scene an event occurred) is a natural extension. Integration with downstream systems (e.g., fine-tuning a VLM on PRISM's selected events as a data-efficient labeling pipeline) would close the loop from pure detection to actionable explanations.

VI. Potential Applications

The ability to localize salient events without training opens a number of downstream applications in robot learning:

Dataset curation. Selecting top-K events per trajectory enables efficient labeling workflows: instead of annotating full trajectories, humans can review 5 representative frames per trajectory, tag them with fine-grained labels (success, semantic error, contact problem, etc.), and use those labels to train downstream models.

Failure diagnosis. In a deployed system, PRISM can run offline over logged trajectories to identify abnormal episodes and pinpoint when failures occurred, enabling both post-hoc analysis and automated alerts to human supervisors.

Multi-embodiment transfer. Because PRISM is modality-agnostic and requires no training, the same weights (in fact, the method is parameter-free) transfer to new robot platforms, new objects, and new task structures, provided they log the same set of sensor types.

VII. Conclusion

PRISM treats salient-event and failure localization for industrial manipulation as a training-free, zero-shot, modality-agnostic signal-detection problem, using topological persistence in place of arbitrary top-K or NMS peak-picking. We show two forms of evidence: on REASSEMBLE's failure-annotated trajectories, once the evaluation is corrected for REASSEMBLE's wide failure-annotation windows, PRISM shows a statistically significant, mathematically grounded advantage over both a uniform baseline and a vision-only baseline, concentrated exactly where it should be — force-dominant, visually silent sub-tasks such as insertion and removal. On a hand-annotated set of successful trajectories spanning a taxonomy of sub-events (contact, insertion/extraction, grasping, stalls), PRISM again outperforms both baselines on precision, recall, and especially on tight ($\leq 1\,\mathrm{s}$) temporal tolerance, validating that the underlying capability is general salient-event localization rather than a failure-specific detector. We position PRISM not as a general failure explainer but as an extensible, multimodal event localizer: a first, quantitative stage that downstream semantic systems (vision-language or otherwise) can be built upon.

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